

Survey on Civil Complaints Management System by Using Machine Learning Techniques

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Abstract: *Today's Government Complaints Registration System is totally Human Operable hence it consumes lot of time to resolve each of those complaint. The proposed system is being developed for the government offices to create a helpful complaint registering platform which will be efficient. The Framework model of civil complaint handling system is done by using sentimental analysis and ML techniques to speed up the process of categorization and prioritization of complaints. The system will analyses the citizen sentiment to prioritize the complaints. Later then the categorization of complaints done by using clustering method and give them priority based upon urgency of each one respectively. The municipality can use this method to identify citizen's needs and estimate their satisfaction.*

Keywords: Classification, Clustering, K-means, Prioritization, Sentiment Analysis

I. INTRODUCTION

In Today's time the citizens need to go to government offices physically to register their complaints. Now days people do not have much time to follow the same procedure to get their complaints registered and also have to wait for long time to get them resolved. The government offices receive numerous complaints each day and hence becomes difficult to sort them and get them resolved on time. Sentiment analysis can useful for building required results as there is huge volume of data which is generated by e-complaints.

In previous years, different algorithms get used for sentiment analysis such as Lexicon Based Sentiment Analysis, Naïve bayes, Support Vector Machine (SVM) etc. Also, for clustering purpose k-means and BA (Bees Algorithm) algorithms were commonly used. As per the study, we noted that using Machine Learning techniques and data mining tools could be very useful for our proposed model.

The main goal of this study is to develop a system for the government offices to create a helpful complaint registering platform which will be efficiently deal with management. Initially different types of complaints would use to be gathered at one place which used to create a lot of ambiguity for the officials. This proposed framework will create a data set from all registered complaints and sort them to reduce the ambiguity.

Our research aims to tackle the current problem by analyzing the complaint's sentiment to prioritize the civil complaints. Later then Categorization complaints to their respective departments will be done by use of clusters and priority will be given based upon urgency of each one respectively. As a result, complaints can get managed within the time. This system will help to ensure the work is getting done and gain the public satisfaction with less human efforts.

II. LITERATURE SURVEY

In 2011, ABD el-Fatah A. Hegazy, et al [5] describes the Complaint Management System oriented by Web-application which can be utilize by public for making complaints about services provided by Government. The system can handle complaints by making record and giving feedback. Results of this process are useful towards generating a reference for finding the needs of citizen from complaints and resolving processes of complaints by agencies in body of any organization.

The combination of lexical-based and machine-learning techniques used in classification method SentBulk. Although lexicon-based sentiment analyzers had been used along the years, researcher's combined well-known techniques with

standard lexicon-based approach to improve the results of the analysis of Facebook messages. On the other hand, it had created different hybrid classifiers by combining our own lexicon-based approach with several machine-learning techniques, to improve the performance of Facebook message sentiment analysis was proposed in 2013 by Alvaro Ortigosa, et al [6].

The work in paper [1] uses the annotations for the phrases to present an extensive analysis of how negators, modals, and degree adverbs impact the sentiment of other words in their particular scope. In this work, we provide an extensive analysis of the impact of different modifiers on sentiment based on reliable fine-grained manual annotations.

In 2019 Xingming Chen, et al [2], have published an effective model for sentiment classification. Without dropping any condition, the proposed model features the sentiment reversing effect of negatively phrased words and the sentiment shifting towards the effect of intensive words. Particularly, this method can solve the domain-specific problem without relying on the external sentiment lexicons.

In the proposed model in paper [3], the data was gathered and pre-processed, then it was transformed into a suitable format for clustering. Prioritization and clustering were done on to three factors, and then estimation of satisfaction which gave superior output. K-means method was 0.299 more than other methods used. Increase in the frequency change domain and recency causes faults. The average of all the responses from the executives are the factor that quantifies the better analysis.

Also, Afify, et al. (2011) proposed that [1], is very associated with the research topic because it presents a brand-new model of e-Complaint web service supported Service-Oriented architecture (SOA). In this model, the researchers have tried to regain the association between citizens and social commonality. The projected model aims to make a Web-based e-Complaint related to the subsidiary lifecycle. The cycle begins with the distribution of assorted services equipped through the subsidiary. Those services were delivered to varied folks in step with their needs. Due to totally different hindrances, those services might not be efficiently applied. Because of that, there has been a must for a system that tracks down and discovers citizens issues and provides them with appropriate feedback. This system will manage grievances through recording them and giving feedback for every raised complaint. The study outcome has been a useful regard to confirm users' requirements from the e-Complaint and also the managing method of this complaint within the core of any firm.

III. LIVE SURVEY

In early 2019, it had been discovered that customer Complaints are thought of very important and vital information that may be used to achieve customers' satisfaction. Ensuing to that, institution of a complaint handling system is necessary for addressing client needs and dissatisfactions' and preventing same issues from reoccurring. Economical managing of complaints is that the ability to be attained and enhanced through training and learning. The "Service" has connected completely different databases from different platforms to retrieve certain data. Service implementation starts with the help of web-application that interface development. Afterward, the "Service" is employed to feature the 3 facilities utilized in the recommended e-complaint web service. These services are considered to accumulate the citizen and workers information and conclude however they're performing. Afterwards, researchers' have implemented the web application, every operation calls the service to restore certain data. Considering this, given paper lays emphasis on the services provided by government in context to management of complaints [4].

Customers are the essential factor in the organization. The business has got to support the clients' preferences and demands for creating the customer loyalty, that make the client still purchases with the actual company. The client might feel discontented with the service once he or she receives the delay of services and that they don't recognize the channel for filing the criticism, and additionally the present complaint handling within the organizations still has the issues. Therefore, we, developers of this project enforced the good complaint Management System (SCMS) consisting of the mobile application, chatbot and internet application, for finding the customer's dissatisfaction issue. What is more, the SCMS has the service for classifying the complaint, then automatically direct to the accountable department, and therefore the service for locating the similar complaint to avoid submitting the duplicate complaint. The test result

shows that this technique is ready to cut back the time and procedures for complaint handling, increase the channel for filing the complaint, and increase the channel for progress news and following the status of the complaint [7].

IV. ALGORITHMIC SURVEY

A. Lexicon-Based Sentiment Analysis

In the new approach, the sentiment is normalized, which allows us to get the intensity of sentiment instead of positive/negative decision. A new evidence-based combining function was developed to boost performance of the algorithm within the cases where a mixed sentiment occurs in a very message. The sentiment was performed on Stanford Twitter test set and IMDB data set. It results that the 2 new functions improve performance of the quality lexicon-based sentiment analysis algorithm. It may well be noticed that the strategy is more appropriate for brief messages like tweets. When applied with long documents the strategy performed significantly better on the sentence than on the document level [8].

B. K-Means

The K-Means clustering algorithm is a partition-based cluster analysis method [9]. According to the algorithm, first selection of k objects as initial cluster centers is done, then calculate the distance between each object and every cluster center and assign it to the nearer cluster, give updated status of the averages of all clusters. Repeat this process till the function of criterion converged.

The K-Means algorithm is incredibly sensitive to the initial cluster centers so the clustering results are going to be very different result from different initial cluster centers. If the data points exist in isolation, in other words, a tiny low amount of data points are far away from data-intensive areas, the calculation of the mean point would be affected, and also the new cluster center may deviate from verity data-intensive area which eventually result in a clustering output results of large deviation.

C. CNN

It generates sentence representation by a convolutional layer with multiple kernels (i.e., size of kernel's should be 3, 4, 5 with 100 feature maps each) and pooling operations. Note that the dropout operation is added to stop over fitting [2].

D. LSTM

The full collection is process as one sequence. Long Short-Term Memory generates the sentence representation by calculating the mean of the entire hidden states of all words. The hidden state size was empirically set to 128[2].

E. Support Vector Machine (SVM)

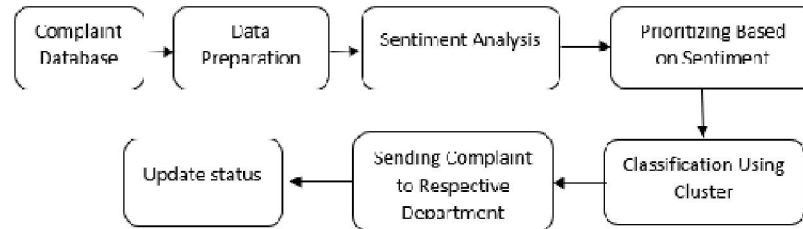
One amongst the classification methods using machine learning (supervised learning) that predicts classes supported models or patterns from the results of the training process. SVM has the advantage of having the ability to spot a separate hyperplane that maximizes margins between two different classes [10]. the subsequent method which will be wont to classify data that can't be classified linearly is by transforming data into high-dimensional feature space in order that it will be separated linearly in space features.

V. PROPOSED SYSTEM

In the proposed mode, there are two main building blocks which are given below:

1. Sentiment Analysis
2. Clustering

Sentiment Analysis is used to check the emotions or feelings of citizens whereas the clustering part is used for classifying the complaints. Sentiment Analysis will spot the negative comments early which will help to work more efficiently. It can be useful for building required results as there is huge volume of data which is generated by e-complaints and on social media



When a Citizen has an issue with certain thing, the next steps need to be followed:

- The Citizen will login to the web portal and writes in his/ her complaint and report the issue.
- All the complaints collected by system in form of data set goes for data pre-processing.
- Data Preparation will clean the data before analysis.
- In preparing the data, the impartial and repetitive data are deleted from the database.
- Sentiment Analysis will spot the negative comments early which will help to work more efficiently.
- Priorities are given to the complaints on the basis of the results achieved by sentimental analysis.
- Then complaints are classified based on the clusters according to the “Frequency” and “Time Interval” by using K-means algorithm.
- The concerned Staff handle the prioritised complaints given to each department and send them to the respective employees.
- After getting complaints resolved, the status will be updated on the web so that citizen can have track of their complaint.

VI. CONCLUSION

With this proposed model, government can analyse citizen’s complaint data to get a clearer knowledge of citizen satisfaction with municipal services performance. Also, they can identify citizen’s service needs and provide important information to government administrators. Through the results of prioritization, working staff will get notified about the region where the work has to be done urgently on different urban-related issues. It can be used to design strategies for timely and proactively enhancing the service infrastructure in a community. As a result, Government funds can get used smartly and effectively. Furthermore, knowing the frequent high demands in specific region, local government can predict emergencies and get prepared for it. We desire to improve descriptive results in offering new perspectives into the needs of citizens.

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